

Tutorial 4. Shading and Photometric Stereo Solutions

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Interactive lab: cv-tutorial-04-lab.chenrui-fan.chatgpt.site

Source code: github.com/Sosekie/Unibe-CVG-CV-Interactive-Labs

Scope. This tutorial practises Lecture 4 on shading, shape from shading, integrability and photometric stereo: relating normals to depth, reading the reflectance map, recovering albedo and normals from several lights, and formulating shape from shading. Q7 applies the Lambertian model to a nearby point light; its part (b) goes beyond the slides. Slide numbers refer to the Lecture 4 slides lec04a. The solution sheet repeats each question before its answer.

Conventions. We use the frame of slides 9-12: x and y span the image plane, and $+z$ points toward the camera. The visible surface is $(x, y, z(x, y))$, seen by an orthographic camera. A light direction $\mathbf{s}$ is the unit vector from the surface point toward the light source. In this tutorial every light is on the camera side of the surface, so $s_3 > 0$. As on slide 9, the normal is written as

$$\mathbf{N} = (p, q, 1)^\top, \quad p = -\frac{\partial z}{\partial x}, \quad q = -\frac{\partial z}{\partial y}, \quad \mathbf{n} = \frac{\mathbf{N}}{\|\mathbf{N}\|_2}.$$

Slides 25 and 31-33 draw the z -axis the other way, pointing away from the camera, so the same normal has third component -1 there. Both drawings are consistent: in each of them, $\nabla z = -(n_1/n_3, n_2/n_3)^\top$. Some texts define $p = \partial z / \partial x$ instead, which reverses the signs of p and q ; check the definition before comparing formulas. If image rows increase downward, convert row indices to y first.

Key concepts.

- *Lambertian shading* (slides 7-8). A distant light with unit direction $\mathbf{s}$ and intensity L gives a point with albedo ρ and unit normal $\mathbf{n}$ the brightness $I = \rho L \max(0, \langle \mathbf{n}, \mathbf{s} \rangle)$. For a point that faces the light, this is the slide formula $I = \rho \langle \mathbf{n}, \mathbf{s} \rangle L$; a point with $\langle \mathbf{n}, \mathbf{s} \rangle \leq 0$ is in attached shadow and has $I = 0$. The viewing direction does not appear.
- *Gradient space* (slide 10). Each orientation that faces the camera is one point (p, q) of the plane $z = 1$.
- *Reflectance map* (slides 10-12). With $\rho = L = 1$, as on slide 8, the brightness of an orientation (p, q) is $R(p, q) = \langle \mathbf{n}, \mathbf{s} \rangle = \cos \theta_i$ if it faces the light, and 0 otherwise. R is largest, $R = 1$, when $\mathbf{n} = \mathbf{s}$ (slide 12). An iso-brightness contour collects all orientations with the same brightness.
- *Integrability*. A field (p, q) is the slope field of a smooth depth map only if $p_y = q_x$.
- *Photometric stereo* (slides 14-20). Camera and object stay fixed while known lights change; each pixel gives a linear system for $\mathbf{g} = \rho \mathbf{n}$.
- *Normal integration* (slides 25-28). Least squares turns a normal map into a depth map, up to an additive constant.
- *Shape from shading* (slides 30-34). One image plus prior constraints, solved by minimising an energy.

Surface normals and shading

Focus: normals and depth slopes, integrability, the reflectance map and the single-image ambiguity.

Q1. Normals, depth and integrability (slides 9, 25-28).

- (a) Use the tangent vectors of $(x, y, z(x, y))$ to show that $\mathbf{N} = (p, q, 1)^\top$ with $p = -z_x$ and $q = -z_y$ is normal to the surface. Express ∇z through a measured unit normal $\mathbf{n}$.
- (b) Compute p, q and $\mathbf{n}$ for the plane $z = 2x - y + 3$. Why is n_1 negative?
- (c) Show that a field (p, q) can come from a smooth depth map only if $p_y = q_x$. Test $(p, q) = (y, x)$ and $(p, q) = (y, 0)$.
- (d) On a pixel grid, write the least-squares problem that recovers z from a normal map with finite differences. Why must one depth value, or the mean depth, be fixed? What goes wrong where $n_3 \rightarrow 0$?

Solution.

(a) The surface point $P(x, y) = (x, y, z(x, y))^\top$ has the tangent vectors

$$P_x = (1, 0, z_x)^\top, \quad P_y = (0, 1, z_y)^\top.$$

A normal is orthogonal to both (slide 9):

$$\mathbf{N}^\top P_x = p + z_x = 0, \quad \mathbf{N}^\top P_y = q + z_y = 0 \quad \implies \quad p = -z_x, \quad q = -z_y.$$

Equivalently, $P_x \times P_y = (-z_x, -z_y, 1)^\top = \mathbf{N}$, which points toward the camera. Normalising gives $\mathbf{n} = (p, q, 1)^\top / \sqrt{1 + p^2 + q^2}$. Because $n_1/n_3 = p$ and $n_2/n_3 = q$,

$$\boxed{\nabla z = (z_x, z_y)^\top = -\left(\frac{n_1}{n_3}, \frac{n_2}{n_3}\right)^\top} \quad (n_3 \neq 0),$$

which is the relation on slide 25.

(b) For $z = 2x - y + 3$, $z_x = 2$ and $z_y = -1$, so $p = -2$, $q = 1$ and

$$\mathbf{N} = (-2, 1, 1)^\top, \quad \mathbf{n} = \frac{1}{\sqrt{6}}(-2, 1, 1)^\top.$$

The plane rises toward $+x$, so it faces the $-x$ side: tilting a surface upward in one direction turns its normal the opposite way. This is why p and z_x have opposite signs. Check: $-(n_1/n_3, n_2/n_3) = -(-2, 1) = (2, -1) = (z_x, z_y)$.

(c) For a twice continuously differentiable depth map, $z_{xy} = z_{yx}$. Hence

$$p_y = -z_{xy} = -z_{yx} = q_x.$$

The minus signs cancel, so the condition $p_y = q_x$ does not depend on the sign convention. For $(p, q) = (y, x)$, $p_y = 1 = q_x$: the field is integrable, and $z_x = -y$, $z_y = -x$ give $z = -xy + C$. For $(p, q) = (y, 0)$, $p_y = 1 \neq 0 = q_x$: no depth map has these slopes. Integration then depends on the path. From $(0, 0)$ to $(1, 1)$, z changes by 0 along the path through $(1, 0)$, but by -1 along the path through $(0, 1)$. Measured normal maps are noisy and in general not exactly integrable; the least-squares solution in (d) returns the depth map whose slopes are closest to the measured ones.

(d) Sample $z_{i,j} = z(ih, jh)$ with pixel spacing h , where i counts along x and j along y , and compute $p_{i,j} = n_1/n_3$ and $q_{i,j} = n_2/n_3$ from the normal map. Forward differences give one equation per pair of neighbours:

$$\frac{z_{i+1,j} - z_{i,j}}{h} = -p_{i,j}, \quad \frac{z_{i,j+1} - z_{i,j}}{h} = -q_{i,j}.$$

Stack all depths into a vector $\mathbf{z}$ and the differences into sparse matrices D_x and D_y :

$$\min_{\mathbf{z}} \left\| \begin{bmatrix} D_x \\ D_y \end{bmatrix} \mathbf{z} + \begin{bmatrix} \mathbf{p} \\ \mathbf{q} \end{bmatrix} \right\|_2^2.$$

There are about twice as many equations as unknowns, so least squares averages noise and inconsistencies. Adding the same constant to every depth leaves every difference unchanged. The constant vector therefore lies in the null space of the stacked matrix, and the normal equations are singular. Fix one value, for example $z_{0,0} = 0$, or require zero mean depth: normals carry no information about absolute depth. Where $n_3 \rightarrow 0$, the normal is perpendicular to the viewing direction (an occluding boundary), and $p, q \rightarrow \infty$. A depth map cannot describe a surface seen edge-on, so such pixels must be excluded or down-weighted; Q6 uses another parametrisation there.

Q2. Reflectance map and the single-image ambiguity (slides 7-8, 10-13). A Lambertian surface with $\rho = 1$ is lit by a distant light of intensity $L = 1$. The light direction is $\mathbf{s} \propto (p_s, q_s, 1)^\top = (1, 0, 1)^\top$.

- (a) Which quantities change the brightness of a Lambertian point? Why is the viewing direction not one of them?
- (b) Write the reflectance map $R(p, q)$. Where is it maximal, and on which line is it zero?
- (c) A pixel has brightness $I = 1/\sqrt{2}$. Show that every orientation on the parabola $q^2 = 2p$ explains this value, and give two clearly different normals on it. What does this mean for recovering shape from one image?

Solution.

(a) The brightness $I = \rho L \langle \mathbf{n}, \mathbf{s} \rangle$ depends on the albedo ρ , the light intensity L , the surface orientation $\mathbf{n}$ and the light direction $\mathbf{s}$. A Lambertian surface reflects light equally in all directions, so its brightness is the same from every viewpoint. Moving the camera changes where a point appears in the image, not how bright it is.

(b) With $\mathbf{n} = (p, q, 1)^\top / \sqrt{p^2 + q^2 + 1}$ and $\mathbf{s} = (1, 0, 1)^\top / \sqrt{2}$ (slide 10),

$$R(p, q) = \frac{pp_s + qq_s + 1}{\sqrt{p^2 + q^2 + 1} \sqrt{p_s^2 + q_s^2 + 1}} = \frac{p + 1}{\sqrt{2} \sqrt{p^2 + q^2 + 1}}$$

for orientations with $p \geq -1$. Orientations with $p < -1$ face away from the light and are in attached shadow, $R = 0$. Since $R = \cos \theta_i \leq 1$, with equality only for $\mathbf{n} = \mathbf{s}$, the maximum is

$$\boxed{R = 1 \text{ at } (p, q) = (p_s, q_s) = (1, 0)},$$

which is the rule *maximum reflectance when $\mathbf{n} = \mathbf{s}$* of slide 12. R is zero on the line $pp_s + qq_s + 1 = 0$, here $p = -1$, where $\theta_i = 90^\circ$.

(c) Setting $R(p, q) = 1/\sqrt{2}$ gives

$$p + 1 = \sqrt{p^2 + q^2 + 1} \quad \implies \quad (p + 1)^2 = p^2 + q^2 + 1 \quad \implies \quad \boxed{q^2 = 2p}.$$

Every orientation on this parabola has the same brightness. For example, $(p, q) = (0, 0)$ gives $\mathbf{n} = (0, 0, 1)^\top$, which faces the camera, while $(p, q) = (2, 2)$ gives $\mathbf{n} = \frac{1}{3}(2, 2, 1)^\top$, which is tilted by about 70.5° . Both satisfy $\langle \mathbf{n}, \mathbf{s} \rangle = 1/\sqrt{2}$. All such normals make the same angle $\theta_i = 45^\circ$ with $\mathbf{s}$, so they lie on a cone around $\mathbf{s}$ (slide 11); the contour is where this cone meets the gradient plane. Brighter values give closed ellipses around $(1, 0)$, and darker values give open hyperbolas, as on slide 12.

One image gives one equation per pixel for two unknowns, p and q . Even with known light and albedo, a single image therefore does not determine the normals (slide 13). Additional images (photometric stereo, Q3-Q5) or additional assumptions (shape from shading, Q6) are needed.

Photometric stereo

Focus: linear recovery of albedo and normals from known lights; rank, least squares and shadows.

Q3. Minimum number of images (slides 14-16). At one surface point, assume a Lambertian material, a fixed camera and object, and distant lights with known directions $\mathbf{s}_k$ and intensities L_k .

- (a) What is the minimum number of images needed to recover the albedo ρ and the unit normal $\mathbf{n}$? Which condition must the lights satisfy?
- (b) Can the three lights $\mathbf{s}_1 = (0, 0, 1)^\top$, $\mathbf{s}_2 = \frac{1}{\sqrt{2}}(1, 0, 1)^\top$ and $\mathbf{s}_3 = \frac{1}{\sqrt{2}}(-1, 0, 1)^\top$ be used? Explain what cannot be recovered.
- (c) Which further assumptions must hold for the linear equations to be valid?

Solution.

- (a) For an unshadowed measurement under light k (slide 15),

$$I_k = \rho L_k \langle \mathbf{n}, \mathbf{s}_k \rangle = (L_k \mathbf{s}_k)^\top \mathbf{g}, \quad \mathbf{g} = \rho \mathbf{n}.$$

The scaled normal $\mathbf{g}$ has three unknown components, equivalent to the albedo and the two angles of the unit normal. Each image gives one linear equation, so **three images** are the minimum. Stacking them gives $\mathbf{I} = S\mathbf{g}$, where row k of S is $L_k \mathbf{s}_k^\top$. The solution is unique if and only if S is invertible (rank 3): the three light directions must not lie in one plane through the origin, and in particular no two may be parallel. Then $\mathbf{g} = S^{-1}\mathbf{I}$, $\rho = \|\mathbf{g}\|_2$ and $\mathbf{n} = \mathbf{g}/\rho$ (slide 16).

(b) No. All three directions have a zero y -component, so they lie in the x - z plane, the second column of S is zero and $\text{rank } S = 2$. The images constrain only g_1 and g_3 : every $\mathbf{g} + t(0, 1, 0)^\top$ explains them equally well. Thus n_2 , and with it ρ , cannot be recovered. Two lights from the same direction, even with different intensities, are another dependent case.

(c) The material is Lambertian. The lights are distant, with calibrated directions and intensities. Camera, object and albedo do not change between images, so each pixel keeps seeing the same surface point. Pixel values are linear in radiance (no gamma correction) and unsaturated. The point is lit in every image used ($\langle \mathbf{n}, \mathbf{s}_k \rangle > 0$, no attached or cast shadow). There is no ambient light or interreflection, or an image taken with all lights off is subtracted first.

Q4. Solving for albedo and normal (slides 15-16). Use the lights $\mathbf{s}_1 = (0, 0, 1)^\top$, $\mathbf{s}_2 = \frac{1}{\sqrt{2}}(1, 0, 1)^\top$ and $\mathbf{s}_3 = \frac{1}{\sqrt{2}}(0, 1, 1)^\top$ with $L_k = 1$. At one pixel the measured linear intensities are

$$\mathbf{I} = \left(\frac{1}{5}, \frac{3\sqrt{2}}{10}, \frac{3\sqrt{2}}{10} \right)^\top.$$

Form and solve the linear system for $\mathbf{g} = \rho \mathbf{n}$. Recover ρ and $\mathbf{n}$, then convert $\mathbf{n}$ to (p, q) and to the depth slopes (z_x, z_y) . Why is $\mathbf{g}$ not a unit vector?

Solution.

With the light directions as rows,

$$S = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}, \quad S\mathbf{g} = \mathbf{I}.$$

Row 1 gives $g_3 = \frac{1}{5}$. Row 2 gives $(g_1 + g_3)/\sqrt{2} = 3\sqrt{2}/10$, so $g_1 + g_3 = \frac{3}{5}$ and $g_1 = \frac{2}{5}$. Row 3 gives $g_2 = \frac{2}{5}$ in the same way. Equivalently, using the inverse matrix,

$$S^{-1} = \begin{bmatrix} -1 & \sqrt{2} & 0 \\ -1 & 0 & \sqrt{2} \\ 1 & 0 & 0 \end{bmatrix}, \quad \mathbf{g} = S^{-1}\mathbf{I} = \begin{bmatrix} 2/5 \\ 2/5 \\ 1/5 \end{bmatrix}, \quad \boxed{\rho = \|\mathbf{g}\|_2 = \frac{3}{5}, \quad \mathbf{n} = \frac{\mathbf{g}}{\rho} = \begin{bmatrix} 2/3 \\ 2/3 \\ 1/3 \end{bmatrix}}.$$

Check: $\rho(\langle \mathbf{n}, \mathbf{s}_1 \rangle, \langle \mathbf{n}, \mathbf{s}_2 \rangle, \langle \mathbf{n}, \mathbf{s}_3 \rangle) = \frac{3}{5}(\frac{1}{3}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) = \mathbf{I}^\top$. All three cosines are positive, so the point is lit in every image.

In gradient space, $(p, q) = (n_1/n_3, n_2/n_3) = (2, 2)$, and the depth slopes are $(z_x, z_y) = -(p, q) = (-2, -2)$: z decreases toward $+x$ and $+y$, and the normal leans toward $+x$ and $+y$.

The linear solve returns albedo times normal. The direction of $\mathbf{g}$ is the normal and its length is the albedo, here $\frac{3}{5}$, so $\mathbf{g}$ is a unit vector only when $\rho = 1$. Photometric stereo gives the normal map and the albedo map; a depth map needs the integration of Q1(d).

Q5. More than three images, shadows and colour (slides 17-20).

- (a) For $m > 3$ lights, write the least-squares estimate of $\mathbf{g}$. When is it unique, and why do extra images help?
- (b) Slide 20 multiplies the k -th equation by its own intensity I_k . Write the resulting system. What happens to a measurement in shadow, and which bad measurements does this step not remove?
- (c) How can colour images be handled (slide 18)?

Solution.

(a) Stack the measurements into $\mathbf{I} \in \mathbb{R}^m$ and the rows $L_k \mathbf{s}_k^\top$ into $S \in \mathbb{R}^{m \times 3}$. Left-multiplying $\mathbf{I} = S\mathbf{g}$ by $S^\top$ and inverting gives (slide 17)

$$\hat{\mathbf{g}} = \arg \min_{\mathbf{g}} \|S\mathbf{g} - \mathbf{I}\|_2^2 = (S^\top S)^{-1} S^\top \mathbf{I} = S^+ \mathbf{I},$$

with the Moore-Penrose pseudo-inverse S^+ . The estimate is unique when S has rank 3, that is, when at least three lit images have linearly independent directions. As before, $\rho = \|\hat{\mathbf{g}}\|_2$ and $\mathbf{n} = \hat{\mathbf{g}}/\rho$. Extra images average out noise. The residual $\|S\hat{\mathbf{g}} - \mathbf{I}\|_2$ reveals measurements that do not fit the model, well-spread lights improve the conditioning of $S^\top S$, and a pixel can still be solved when a few of its measurements must be discarded.

(b) Multiplying equation k by I_k gives (slide 20)

$$I_k^2 = I_k L_k \mathbf{s}_k^\top \mathbf{g}, \quad k = 1, \dots, m, \quad \text{that is,} \quad D\mathbf{I} = DS\mathbf{g}, \quad D = \text{diag}(I_1, \dots, I_m),$$

with the least-squares solution $\hat{\mathbf{g}} = (S^\top D^2 S)^{-1} S^\top D^2 \mathbf{I}$. A shadowed measurement has $I_k \approx 0$, so its row becomes $0 = 0$ and no longer affects the solution (slide 19). Without this step, an attached shadow would impose $(L_k \mathbf{s}_k)^\top \mathbf{g} = 0$, although the linear model predicts a negative value there. At least three independent lit equations must remain. The weighting gives bright measurements more influence. It therefore does not remove saturated pixels or specular highlights: they receive even larger weights and must be detected and discarded first.

(c) Each colour channel $c \in \{R, G, B\}$ gives its own system $\mathbf{I}_c = S\mathbf{g}_c$ with $\mathbf{g}_c = \rho_c \mathbf{n}$: the channels share the normal but have different albedos (slide 18). A simple solution is to estimate $\mathbf{n}$ from one channel, or from a grey-level image, and then fit each albedo with $\mathbf{n}$ fixed:

$$\rho_c = \frac{(S\mathbf{n})^\top \mathbf{I}_c}{\|S\mathbf{n}\|_2^2}.$$

Shape from shading

Focus: one image, regularisation, boundary conditions and iterative minimisation.

Q6. Shape from shading as energy minimisation (slides 30-34). Only one image $I(x, y)$ is available, with $\rho = L = 1$ and a known light direction $\mathbf{s}$.

- (a) Write an energy for the unknown field (p, q) with a data term based on $R(p, q)$, a smoothness term and the integrability penalty from Q1. What does each term do? What happens if the smoothness weight is very small or very large?
- (b) Why do slides 31-33 replace (p, q) by (f, g) ? What is known about $\mathbf{n}$ on the occluding boundary (slide 32), and why does it help?
- (c) Slide 33 minimises $E[f, g] = \iint (I - R(f, g))^2 dx dy + \lambda(\|\nabla f\|^2 + \|\nabla g\|^2) dx dy$. Derive the gradient equation for f . Then use the approximation of slide 34, $-\Delta f_{i,j} \approx 2(f_{i,j}^{t+1} - \bar{f}_{i,j}^t)$, where $\bar{f}_{i,j}^t$ is the average of the four neighbours, to obtain an iterative update.

Solution.

(a) One possible energy is

$$E[p, q] = \underbrace{\iint (I - R(p, q))^2 dx dy}_{\text{data fidelity}} + \underbrace{\lambda \iint (\|\nabla p\|^2 + \|\nabla q\|^2) dx dy}_{\text{smoothness}} + \underbrace{\mu \iint (p_y - q_x)^2 dx dy}_{\text{integrability}}$$

with weights $\lambda, \mu > 0$. The data term makes the rendered brightness match the image, but at each pixel it only restricts (p, q) to an iso-brightness contour (Q2). The smoothness term asks neighbouring normals to be similar and selects one slowly varying solution among the many candidates; this is the regularisation of slide 30. The integrability term asks the field to be the slope field of a surface (Q1), so that it can be integrated into depth; it couples p and q across pixels. If λ is very small, the problem remains ambiguous and noise dominates. If λ is very large, the shape is over-smoothed and no longer reproduces the image. The energy on slide 33 uses the first two terms, written in (f, g) ; the integrability term is an optional addition. After minimisation, the depth follows from the integration of Q1(d).

(b) The gradient-space coordinates $(p, q) = (n_1/n_3, n_2/n_3)$ become infinite when $n_3 \rightarrow 0$, that is, when the angle θ between $\mathbf{n}$ and the viewing direction reaches $\pi/2$ (slide 31; this θ differs from the incidence angle θ_i of slide 10). This happens exactly on the occluding boundary. The stereographic coordinates of slide 31,

$$f = \frac{2p}{1 + \sqrt{1 + p^2 + q^2}}, \quad g = \frac{2q}{1 + \sqrt{1 + p^2 + q^2}}, \quad \text{that is,} \quad (f, g) = \frac{2(n_1, n_2)}{1 + n_3},$$

remain finite there: the boundary maps to the circle $f^2 + g^2 = 4$. In the frame of this sheet, $\mathbf{n} = (4f, 4g, 4 - f^2 - g^2)^\top / (4 + f^2 + g^2)$. Slides 31-33 use the opposite z -axis, as slide 25 does. There the third components of $\mathbf{n}$ and $\mathbf{s}$ both change sign, so $R = \langle \mathbf{n}, \mathbf{s} \rangle$ is the same, and the data term on slide 33 is exactly this R .

On the occluding boundary the line of sight grazes the surface, so $\mathbf{n} \perp \mathbf{v}$ for the viewing direction $\mathbf{v}$. The normal is also orthogonal to the tangent $\mathbf{e}$ of the boundary curve. Hence $\mathbf{n} = \mathbf{e} \times \mathbf{v}$, oriented outward (slide 32). These normals are known from the silhouette alone. They serve as boundary conditions, which can be imposed in (f, g) as finite values. Because they point outward, they also select the convex shape where a convex and a concave shape would produce the same shading (compare the bump and the hole on slide 3).

(c) The gradient equation (Euler-Lagrange equation) of $E[f, g]$ with respect to f is

$$-2(I - R) \frac{\partial R}{\partial f} - 2\lambda \Delta f = 0 \quad \Longleftrightarrow \quad (I - R) \frac{\partial R}{\partial f} + \lambda \Delta f = 0,$$

and the equation for g is analogous. Slide 34 applies the time-explicit approximation $-\Delta f_{i,j} \approx 2(f_{i,j}^{t+1} - \bar{f}_{i,j}^t)$ only to the regularisation part and evaluates the data part at iteration t :

$$\boxed{f_{i,j}^{t+1} = \bar{f}_{i,j}^t + \frac{1}{2\lambda} (I_{i,j} - R_{i,j}^t) \left(\frac{\partial R}{\partial f} \right)_{i,j}^t}, \quad R_{i,j}^t = R(f_{i,j}^t, g_{i,j}^t),$$

where $(\partial R / \partial f)_{i,j}^t$ is also evaluated at $(f_{i,j}^t, g_{i,j}^t)$. The update for g is the same with $\partial R / \partial g$. Each step averages the neighbours (smoothness) and then moves f in the direction that reduces the brightness error (data fidelity). Boundary pixels keep their known values, and the iteration stops when the changes become small. The constant 2 in the approximation depends on how the smoothness term is discretised; another choice would only rescale λ .

Beyond a distant light

Focus: a nearby point light changes the light direction across a flat surface.

Q7. A nearby point light (slides 7-8, 12). The plane $z = 0$ is Lambertian with $\rho = 1$ and is imaged by an orthographic camera, so pixel (x, y) sees the point $\mathbf{X} = (x, y, 0)^\top$. A point light with $L = 1$ is at $\mathbf{A} = (5, 5, 10)^\top$. Because the light is close, use a different unit direction at every point, $\mathbf{s}(x, y) = (\mathbf{A} - \mathbf{X}) / \|\mathbf{A} - \mathbf{X}\|_2$, and keep L constant.

- Derive the brightness $B(x, y)$ and find the brightest point. Relate the answer to slide 12.
- Extension.* The light of a real point source also falls off as $1/\|\mathbf{A} - \mathbf{X}\|_2^2$. Does this change the brightest point? Compare $B(0, 0)/B(5, 5)$ with and without this factor.
- What would the distant-light model predict for this plane?

Solution.

(a) The normal is $\mathbf{n} = (0, 0, 1)^\top$ everywhere. With

$$\mathbf{A} - \mathbf{X} = (5 - x, 5 - y, 10)^\top, \quad d(x, y) = \|\mathbf{A} - \mathbf{X}\|_2 = \sqrt{(x - 5)^2 + (y - 5)^2 + 100},$$

the brightness is

$$B(x, y) = \rho L \langle \mathbf{n}, \mathbf{s}(x, y) \rangle = \frac{10}{d(x, y)} = \frac{10}{\sqrt{(x - 5)^2 + (y - 5)^2 + 100}}.$$

Since $d \geq 10$, with equality only at $(x, y) = (5, 5)$, we have $B \leq 1$, and

$$\boxed{\text{brightest point } (5, 5, 0)^\top, \quad B = 1}.$$

It lies directly below the light, where $\mathbf{s} = (0, 0, 1)^\top = \mathbf{n}$. This is the rule *maximum reflectance when $\mathbf{n} = \mathbf{s}$* of slide 12, applied to a light whose direction changes over the plane.

(b) With the fall-off,

$$B(x, y) = \frac{10}{d(x, y)^3} = \frac{10}{((x - 5)^2 + (y - 5)^2 + 100)^{3/2}}.$$

Both the cosine factor $10/d$ and the fall-off $1/d^2$ are largest where d is smallest, so the brightest point does not change. The fall-off increases the contrast: $B(0, 0)/B(5, 5) = 10/\sqrt{150} \approx 0.82$ without it and $(100/150)^{3/2} \approx 0.54$ with it.

(c) A distant light has the same direction $\mathbf{s}$ at every point, so $B = \rho L \langle \mathbf{n}, \mathbf{s} \rangle$ is constant on the plane and no point is brightest. Treating a nearby light as distant would explain the real brightness variation by changing normals, which distorts photometric-stereo results. The orthographic camera only maps the point $(x, y, 0)$ to the pixel (x, y) ; it does not change which point is brightest.